

VENKATRAMAN R

Data Science & AI Engineer | Chennai, India

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Professional Summary

Fresher Data Science & AI Engineer with 7 end-to-end deployed projects across GenAI, fraud detection ML, deep learning forecasting, Power BI dashboards, and automated ETL pipelines. Built **AskMyData** — a live LangChain + FAISS RAG application that queries any dataset in plain English via auto-generated pandas or SQL. Fraud models achieve **AUC-ROC 0.97** with SHAP explainability. Seeking entry-level Data Science, ML Engineering, or AI Engineering roles.

Technical Skills

Languages	Python, SQL
GenAI / LLMs	LangChain (LCEL), FAISS, RAG, OpenAI API, Groq Llama-3.3-70b, HuggingFace, Prompt Engineering
ML / AI	XGBoost, Random Forest, SMOTE, SHAP, Optuna, Scikit-Learn, TensorFlow/Keras, LSTM, Cross-Validation
Data / BI	Pandas, NumPy, Power BI (DAX), Plotly, Streamlit, Parquet, SQLite, Matplotlib, Seaborn, Power Query
Tools	Git, GitHub, Jupyter, VS Code, Google Colab, Excel (Advanced — PivotTables, Slicers, Named Ranges)

Projects

AskMyData — GenAI Dataset Q&A App · *LangChain · FAISS · Groq · OpenAI · Streamlit* Live Demo | GitHub

- LangChain LCEL agent auto-selects pandas or SQL; FAISS RAG retrieves per-column schema context; `ConversationBufferWindowMemory` for multi-turn follow-ups; supports CSV, Excel, JSON, SQLite
- Secure sandbox (whitelist-only builtins, DataFrame copy, auto-retry on failure); Plotly charts auto-generated per result; deployed on Streamlit Cloud

FraudGuard — Mobile Transaction Fraud Detection · *XGBoost · SMOTE · SHAP · Optuna · Streamlit* Deployed | GitHub

- XGBoost + Optuna HPO (100 trials) + SMOTE; F2-score threshold optimisation for fraud cost asymmetry; SHAP per-transaction explainability for compliance audit trails
- AUC-ROC 0.97 · F1 0.92** · Precision 0.93 · Recall 0.91; single + bulk CSV batch prediction

BankShield — Bank Account Fraud Detection · *Random Forest · SMOTE · Optuna · Streamlit* Live Demo | GitHub

- End-to-end: EDA → feature engineering → RF + SMOTE → threshold tuning → Joblib → Streamlit (single + batch CSV)
- AUC-ROC 0.97 · Precision 0.94** · Recall 0.89; transaction velocity more predictive than amount

SalesPulse — Power BI Sales Intelligence Dashboard · *Power BI · DAX · Power Query · Star Schema* GitHub

- 44 custom DAX measures (YoY, MTD/QTD/YTD, customer LTV, profit margin); star schema; executive + regional + product + customer segmentation drill-throughs on \$12.63M Global Superstore data

SalesPulse Forecaster — LSTM 12-Month Sales Forecasting · *TensorFlow · Keras · LSTM · SQLite · Streamlit* GitHub

- 2-layer LSTM (128→64) with MinMaxScaler; recursive 12-month forecast with confidence bands; SQLite pipeline; Streamlit upload → forecast → export UI

DataHunt — Automated ETL Pipeline · *Python · Pandas · Parquet · openpyxl · logging* GitHub

- Modular pipeline (Extractor → Cleaner → Storage → Reporter); outputs Parquet + Excel simultaneously; zero hardcoded logic — all strategies are configurable parameters; modules independently swappable

Vehicle Maintenance Dashboard — Excel Fleet Analytics · *Excel · PivotTables · Slicers · Conditional Formatting* GitHub

- 50,000-vehicle dashboard: KPIs, regional breakdown, service scheduling, cost trends; no macros; identified 18% of vehicles accounting for 67% of total maintenance cost (Pareto pattern)

Education

Professional Certification — Data Science with Artificial Intelligence Jun 2026

Boston Institute of Analytics · Current · Most Relevant Qualification

B.E. Automobile Engineering 2021 – 2024

VISTAS — Vels Institute of Science, Technology & Advanced Studies · CGPA 7.25

Certification Detail

Data Science with AI — Boston Institute of Analytics 2025 – 2026

Curriculum — Python, ML (XGBoost, RF, SVM, Neural Networks), Deep Learning, NLP, Generative AI & RAG, Power BI, SQL, Statistics, EDA. Capstone projects: FraudGuard (BIA Flagship) & BankShield. Focus areas: LangChain & FAISS, RAG, Optuna tuning, SHAP explainability, LSTM forecasting, ETL design, Power BI DAX.